

System on Programmable Chip Design

Lecture: Introduction to SOPC Builder

Lecturer: Dr. Nguyen Anh Vinh

Ref: Truong Quang Vinh, IT-SoC Lab

❖ Class content

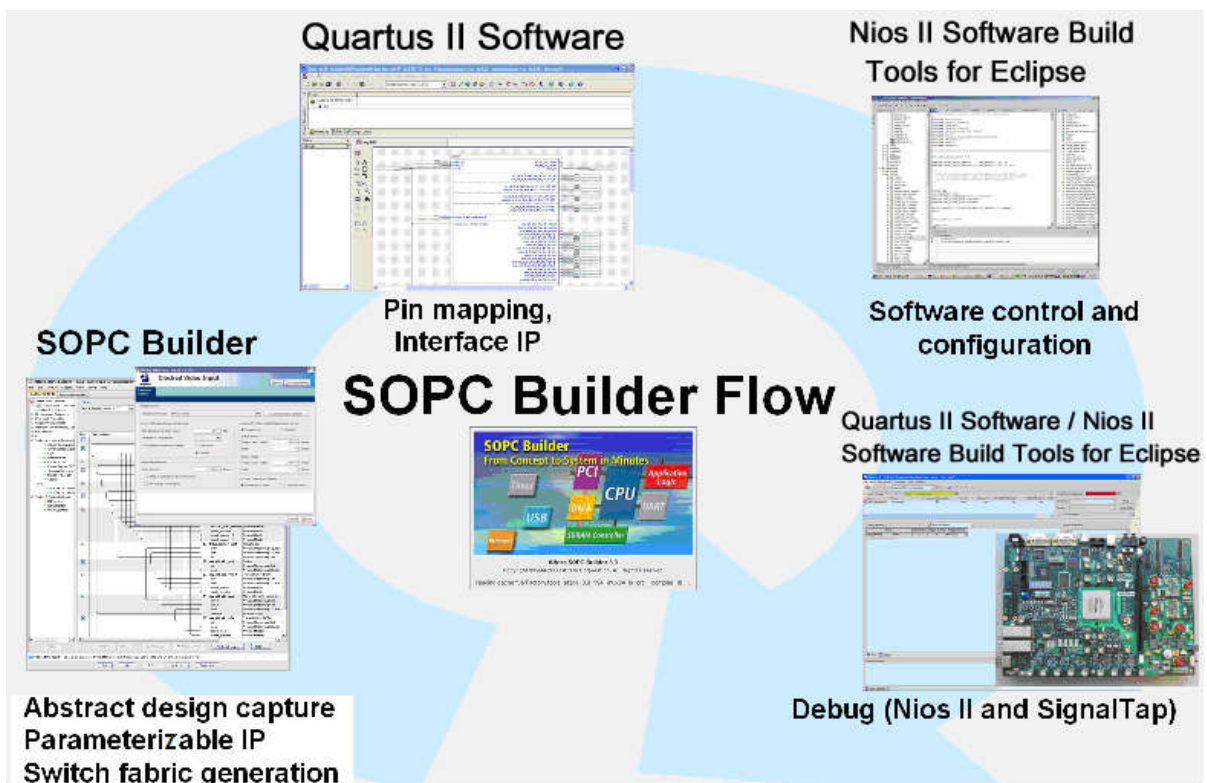
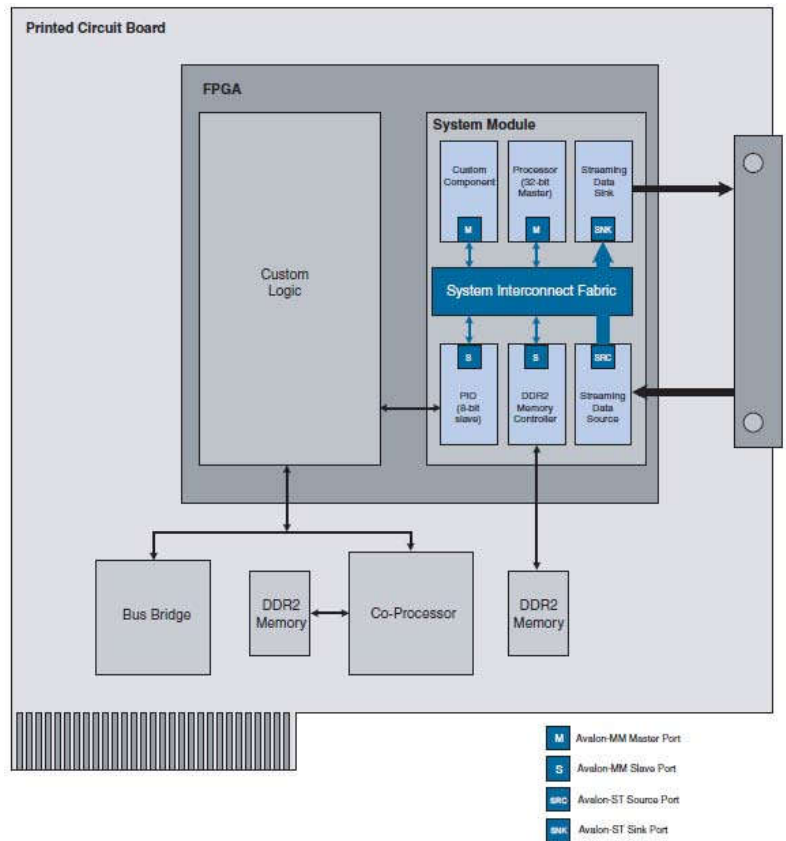
- SOPC system design flow
- Avalon bus system architecture
- SOPC component editor

❖ Required documents

- SOPC Builder User Guide
(http://www.altera.com/literature/ug/ug_sopc_builder.pdf)

❖ SOPC Builder system includes:

- System module
- Custom logic
- Processor
- Memory controller



1. **Package** your component for SOPC Builder using the Component Editor.
2. **Simulate** at the unit-level.
3. **Complete** the SOPC Builder design by adding other components, specifying interrupts, clocks, resets, and addresses.
4. **Generate** the SOPC Builder system.
5. **Perform** system level simulation.
6. **Constrain** and **compile** the design.
7. **Download** the design to an Altera device.
8. **Test** in hardware.

❖ The SoC architecture on SOPC Builder

The screenshot displays the SOPC Builder Target window. The 'Device Family' is set to 'Cyclone III'. The 'Clock Settings' table lists the following:

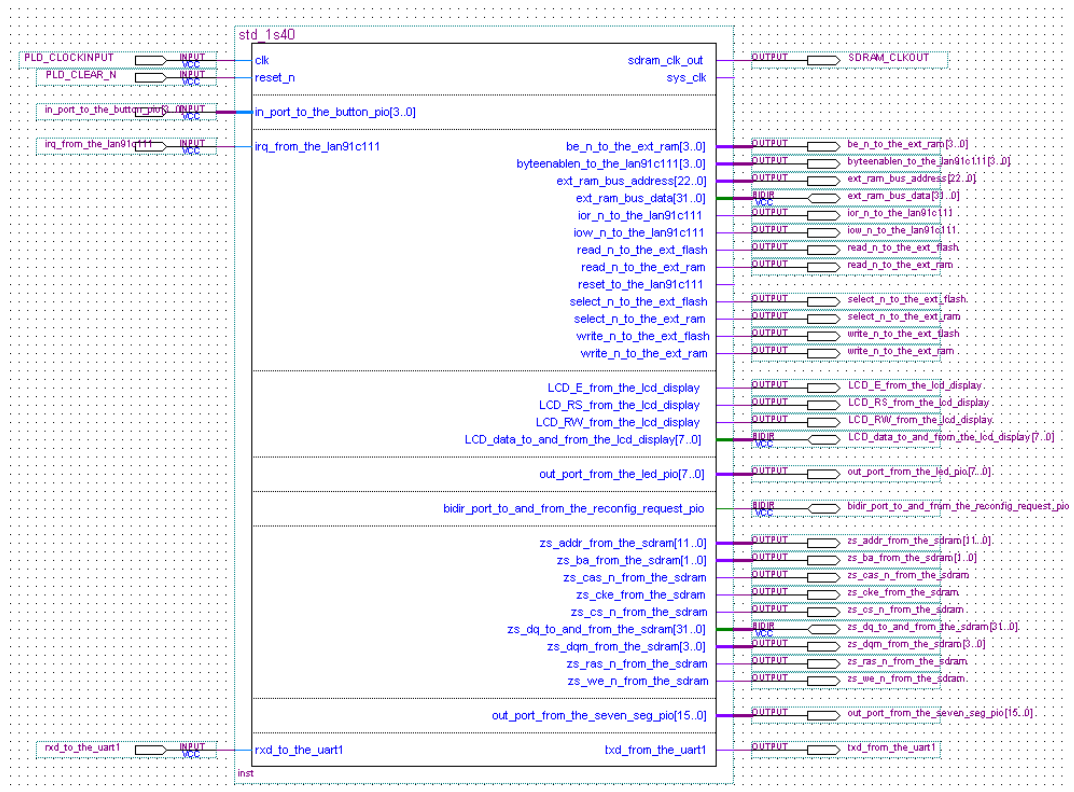
Name	Source	MHz
altmemddr_sysclk	altmemddr.sysclk	75.0
altmemddr_auxfull	altmemddr.auxfull	150.0
altmemddr_auxhalf	altmemddr.auxhalf	75.0
clk_100	External	100.0

The main component list table is as follows:

Use	Conn...	Module Name	Description	Clock	Base	End	IRQ
☒		cpu	Nios II Processor				
☒		instruction_master	Avalon Memory Mapped Master	clk_100			
☒		data_master	Avalon Memory Mapped Master	clk_100			
☒		jtag_debug_module	Avalon Memory Mapped Slave	clk_100			
☒		onchip_mem	On-Chip Memory (RAM or ROM)	clk_100			
☒		jtag_uart_0	JTAG UART	clk_100			
☒		dvi_output_i2c_master	OpenCores I2C Master	clk_100			
☒		quad_video_i2c_mas...	OpenCores I2C Master	clk_100			
☒		my_alt_vip_cti	Clocked Video Input	clk_100			
☒		my_alt_vip_clip	Clipper	clk_100			
☒		my_alt_vip_cpr	Color Plane Sequencer	clk_100			
☒		my_alt_vip_crs	Chroma Resampler	clk_100			
☒		my_alt_vip_csc	CSC	clk_100			
☒		my_alt_vip_dil	Deinterlacer	multiple			
☒		my_alt_vip_clip_1	Clipper	clk_100			
☒		my_alt_vip_scl	Scaler	clk_100			
☒		my_alt_vip_vfb	Frame Buffer	multiple			
☒		my_alt_vip_tpg	Test Pattern Generator	clk_100			
☒		my_alt_vip_mix	Alpha Blending Mixer	clk_100			
☒		my_alt_vip_its	Clocked Video Output	clk_100			
☒		pipeline_bridge	Avalon-MM Pipeline Bridge	altmemddr...			
☒		pipeline_bridge_1	Avalon-MM Pipeline Bridge	altmemddr...			
☒		altmemddr	DDR2 SDRAM High Performance Contr...	phy_refclk			

Arrows point to specific features in the interface:

- Bus connection:** Points to the 'Conn...' column.
- Module name:** Points to the 'Module Name' column.
- Clock signals:** Points to the 'Clock' column.
- Address map:** Points to the 'Base' and 'End' columns.

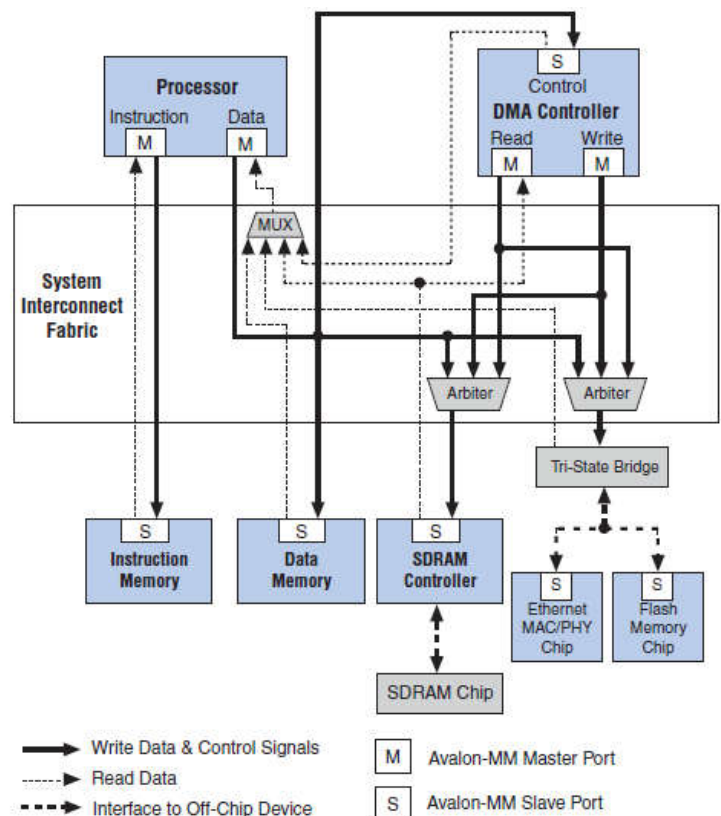


No.	File	Description
1	.qpf	Quartus II Project File
2	.qsf	Quartus II Settings File
3	.v .vhd	HDL design file
4	.sdc	S ynopsis D esign C onstraints File
5	.bsf	B lock D iagram F ile
6	.sopcinfo	SOPC information file
7	.sopc	SOPC Builder Design File
8	.ptf	SOPC Builder System File
	.qip	Quartus II IP File
9	.sof	SRAM Object File

❖ Avalon memory mapped (MM) interface supports:

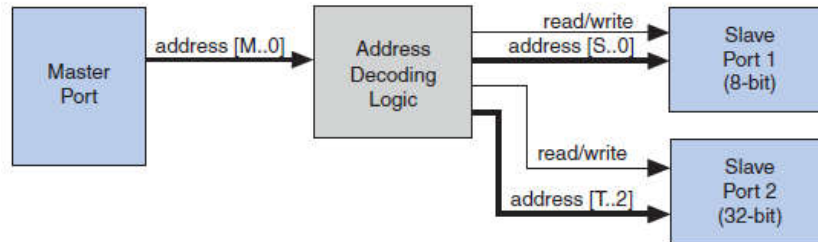
- Any number of master and slave components
- On-chip components
- Off-chip devices
- Components with difference
- Components using multiple Avalon-MM ports.

- ❖ Avalon-MM supports multiple slaves
- ❖ Avalon-MM interfaces are suitable for memories or embedded peripherals
- ❖ Generating system interconnect fabric is SOPC Builder's primary purpose.



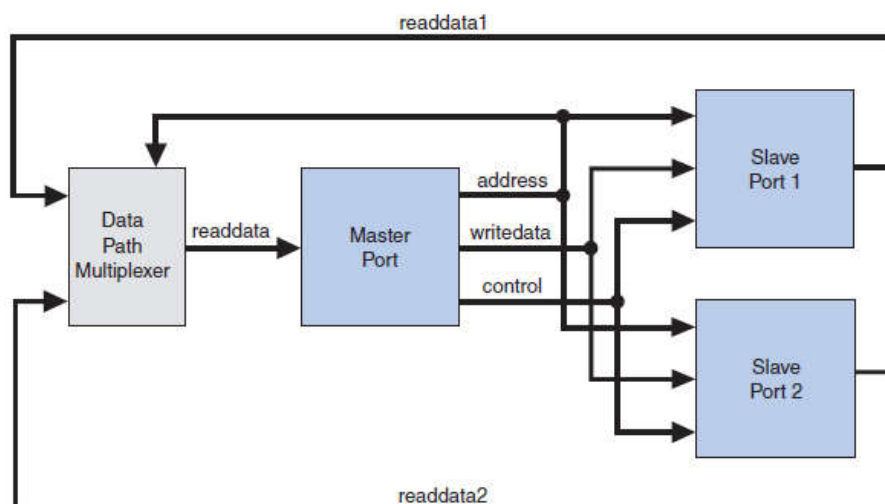
❖ Address decoding supports:

- Select a slave whenever it is being addressed by a d master

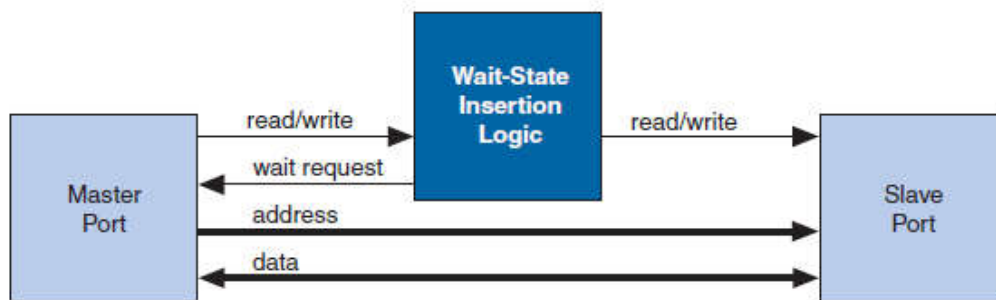


Module Name	Description	Base	End	IRQ
cpu	Nios II Proces...			
instruction_master	Master port			
data_master	Master port			
jtag_debug_mod...	Slave port	0x02120000	0x021207FF	
+ ext_flash	Flash Memory...	0x00000000	0x007FFFFFFF	
+ ext_ram	IDT71V416 S...	0x02000000	0x020FFFFFFF	
+ ext_ram_bus	Avalon Tri-St...			
+ button_pio	PIO (Parallel I/O)	0x02120860	0x0212086F	2
+ high_res_timer	Interval timer	0x02120820	0x0212083F	3

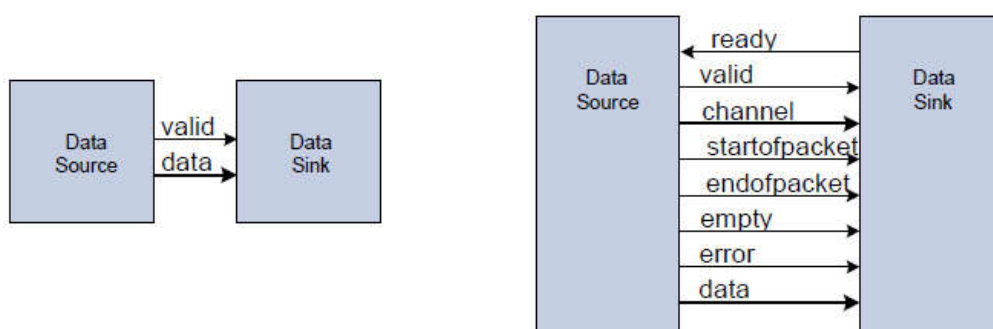
❖ SOPC Builder generates separate datapath multiplexing logic for every master in the system.

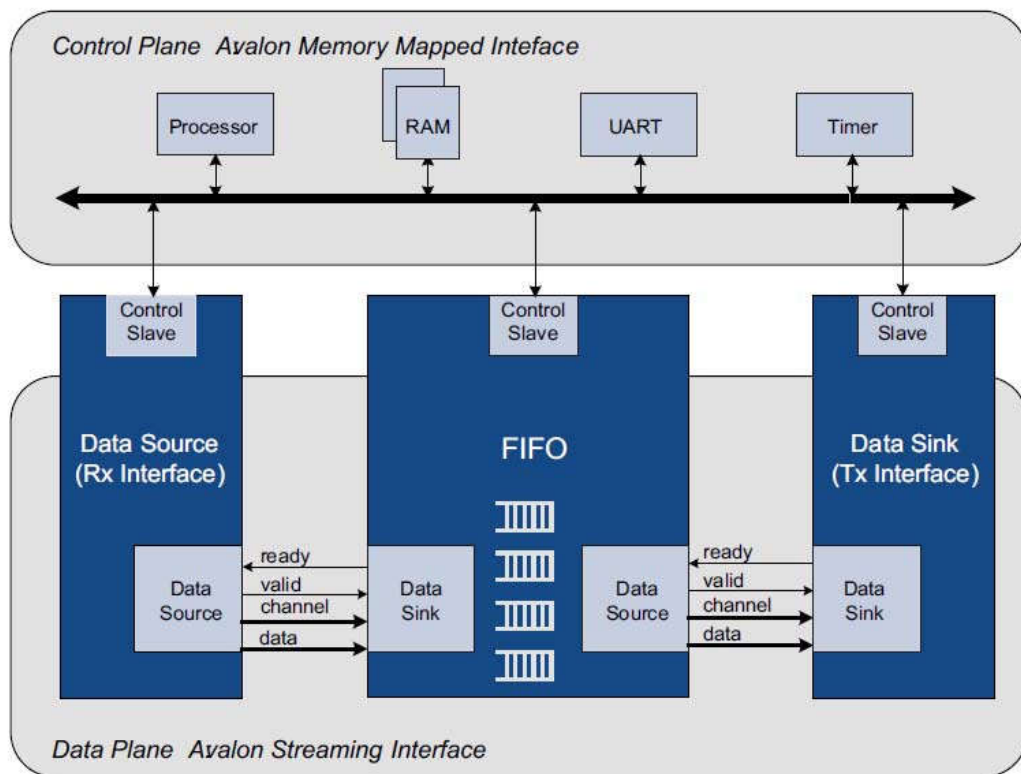


- ❖ Wait states extend the duration of a transfer by one or more cycles.
- ❖ Wait state insertion logic accommodates the timing needs of each slave, and causes the master to wait until the slave can proceed.
- ❖ SOPC Builder generates wait state insertion logic based on the properties of all slaves in the system.

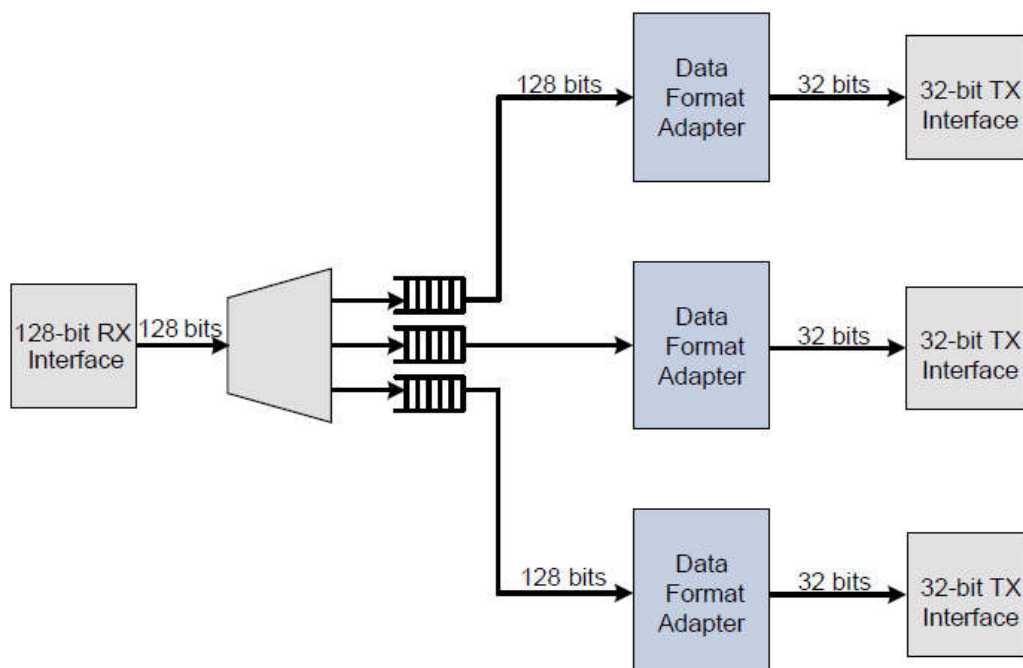


- ❖ Avalon Stream (ST) interface is to connect point-to-point between the two components
- ❖ Synchronous to the rising edge of the associated clock
- ❖ All outputs from the source interface, including the data, channel, and error signals, must be registered on the rising edge of the clock.
- ❖ Registers are not required for inputs at the sink interface.





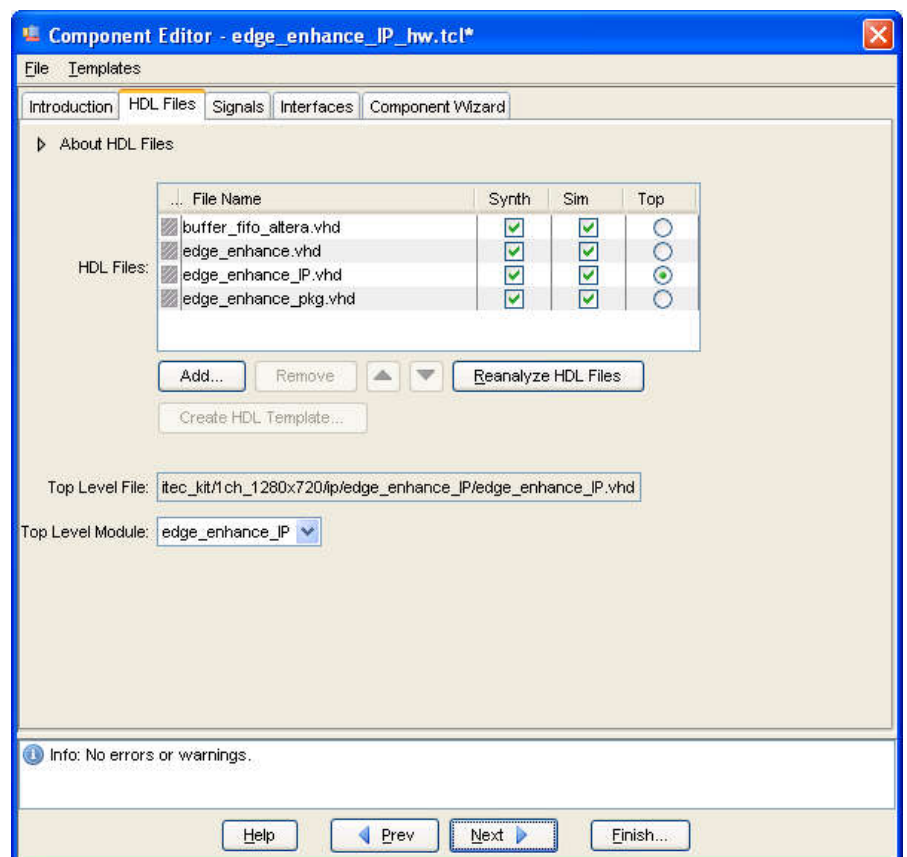
- ❖ The data format adapter allows to connect interfaces that have different values for the parameters



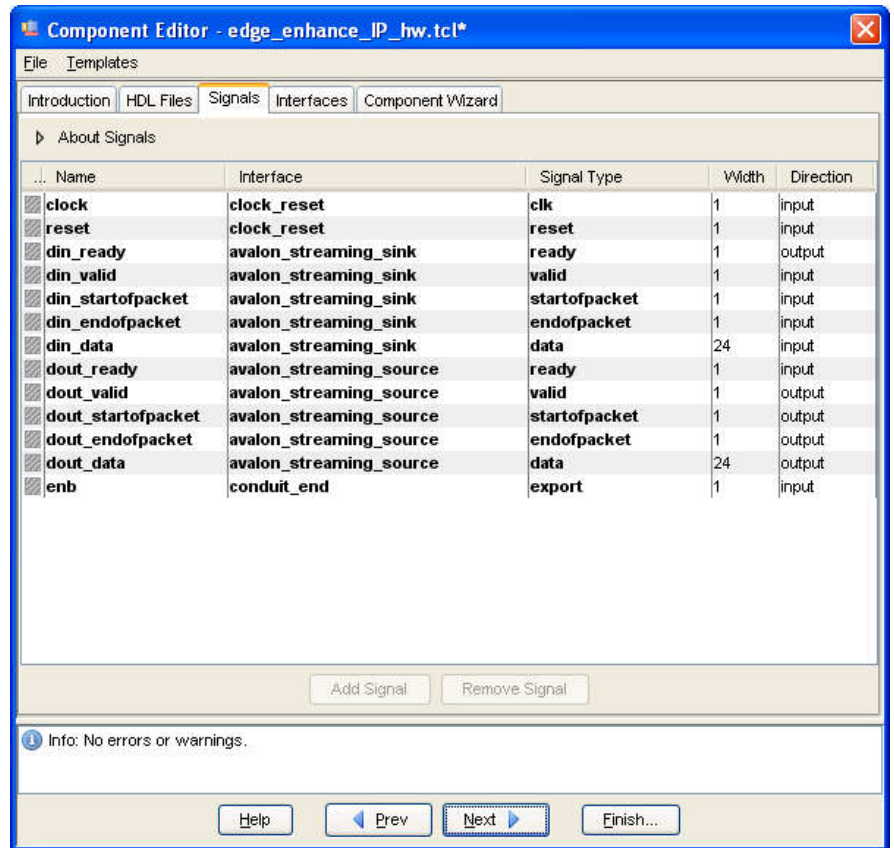
❖ The component editor supports interfaces:

- Avalon-MM master and slave
- Avalon-ST source and sink
- Avalon-MM tristate slave
- Interrupt sender and receiver
- Clock input and output
- Nios II custom instruction master and slave interfaces
- Conduit (for exporting signals to the top level)

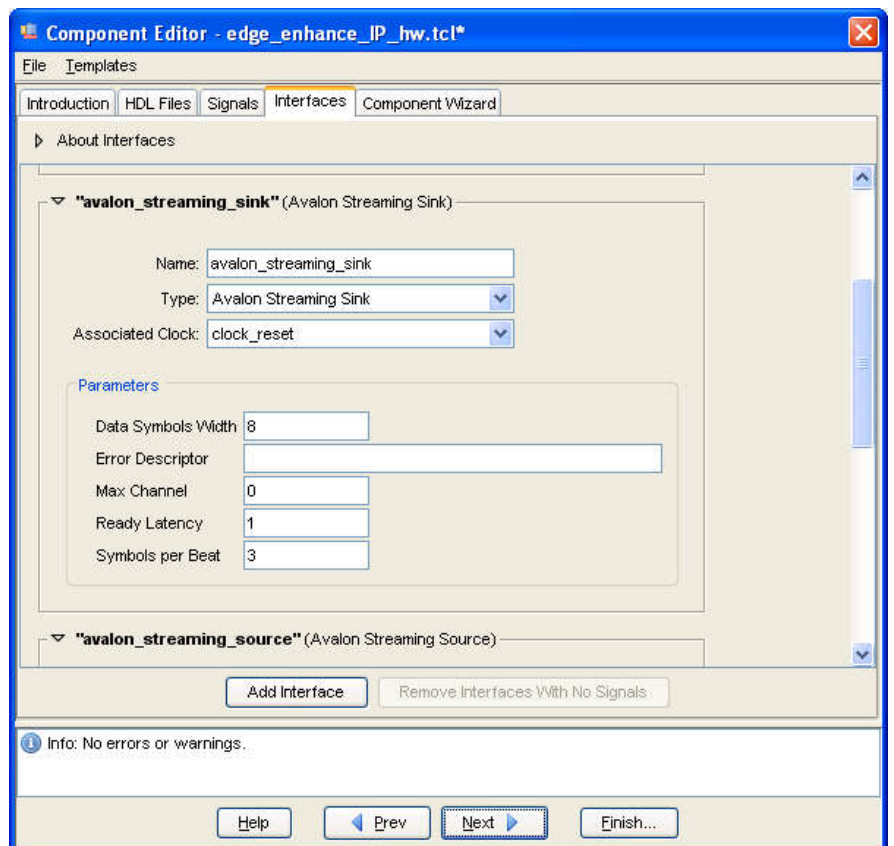
- ❖ Select the HDL files
- ❖ Assign the top block design



❖ Assign types of signals



❖ Assign parameters for Avalon interface

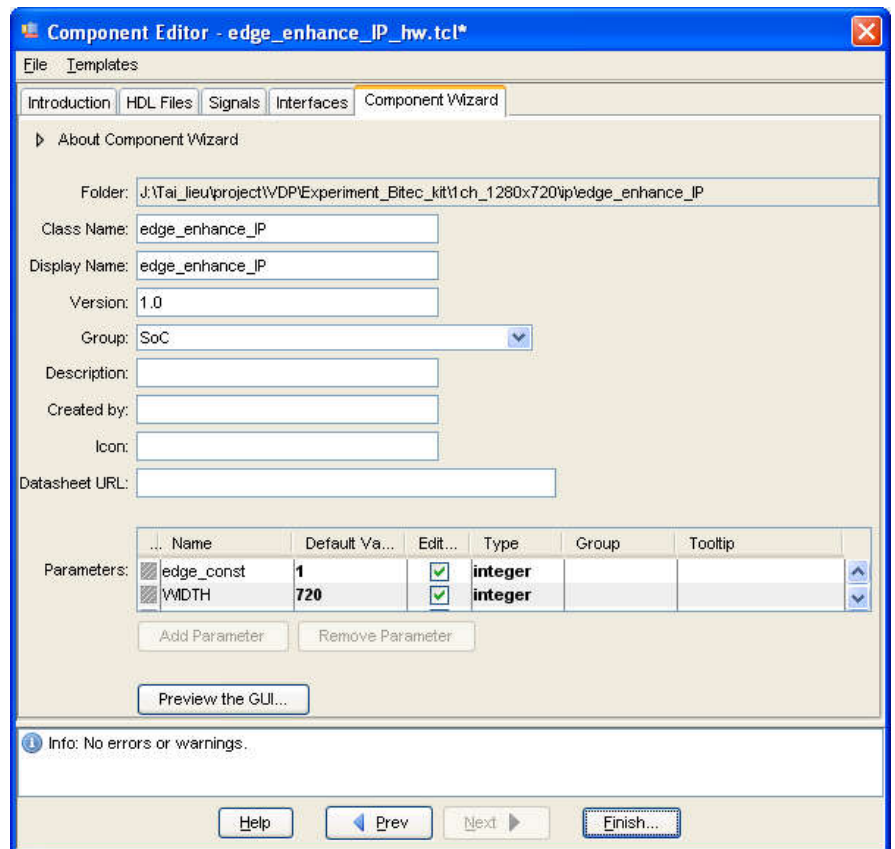


❖ Define

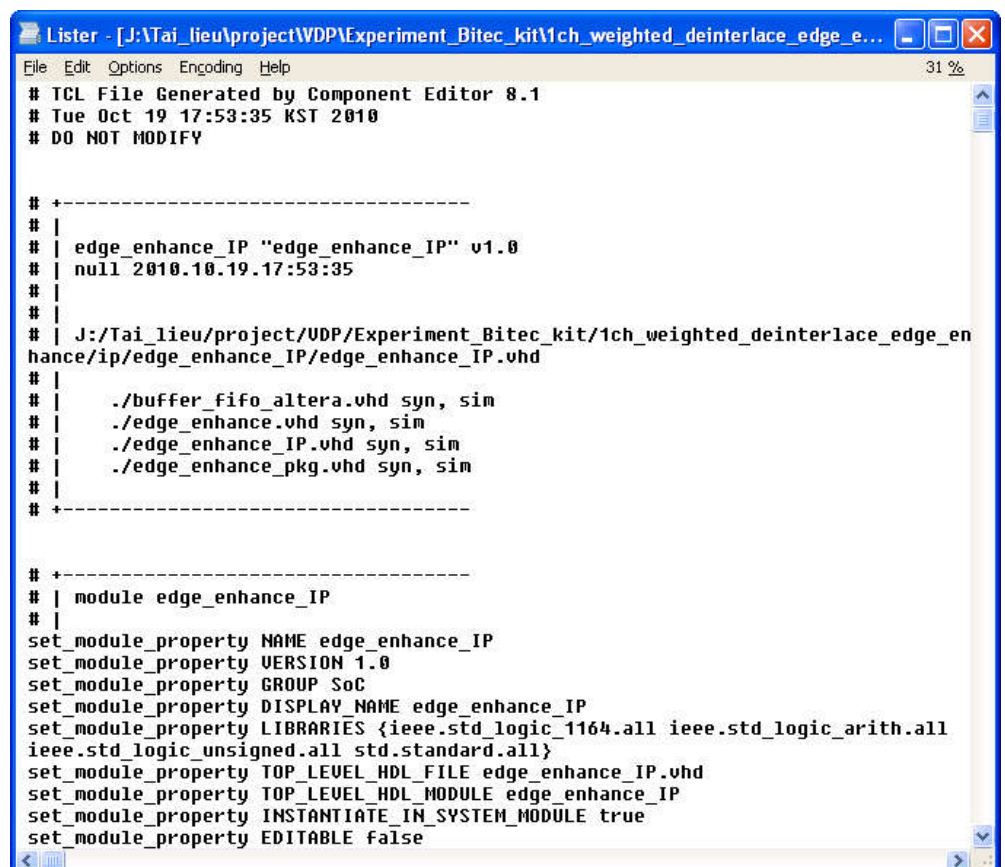
- Class name
- Group name
- Description
- Default parameters

❖ Click Finish to generate component

❖ Output file is **_hw.tcl** file



❖ **_hw.tcl** file: describes the SOPC Builder related characteristics, such as interface behaviors.

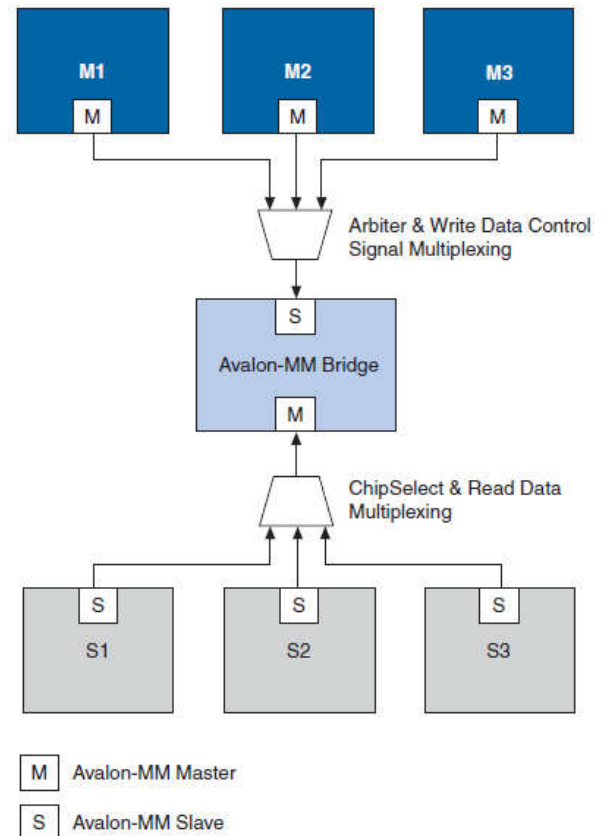


❖ Functions:

- to control the topology of the generated SOPC Builder system.
- By inserting Avalon-MM bridges between masters and slaves, the interconnect is more efficient, and more optimal.
- to separate components in different clock domains

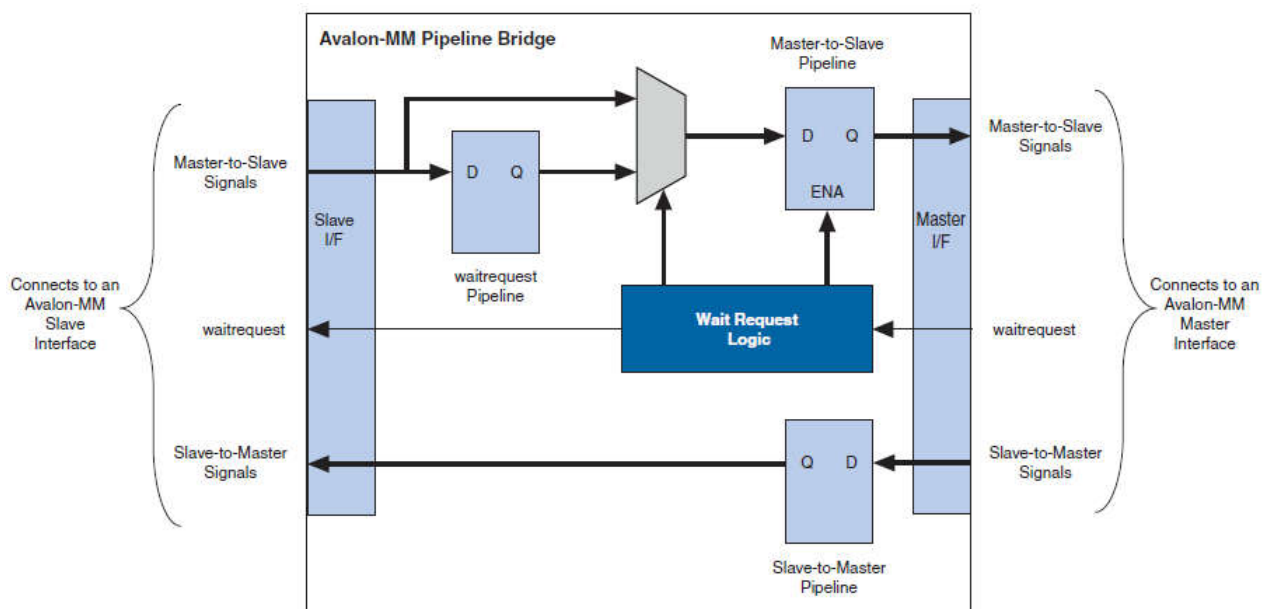
❖ IF there is no bridge between master-slave

- Maximum parallelism is generated (all master can drive transaction to all slaves)
- The system topology is complex, and the performance is slow down



❖ Avalon-MM pipeline bridge

- Insert register in the path between masters and slaves
- Reduce delay at critical timing path



❖ The Avalon-MM clock-crossing bridge

- Allow to connect Avalon-MM master and slave that operate in different clock domains
- Can pipeline multiple read and write transfers

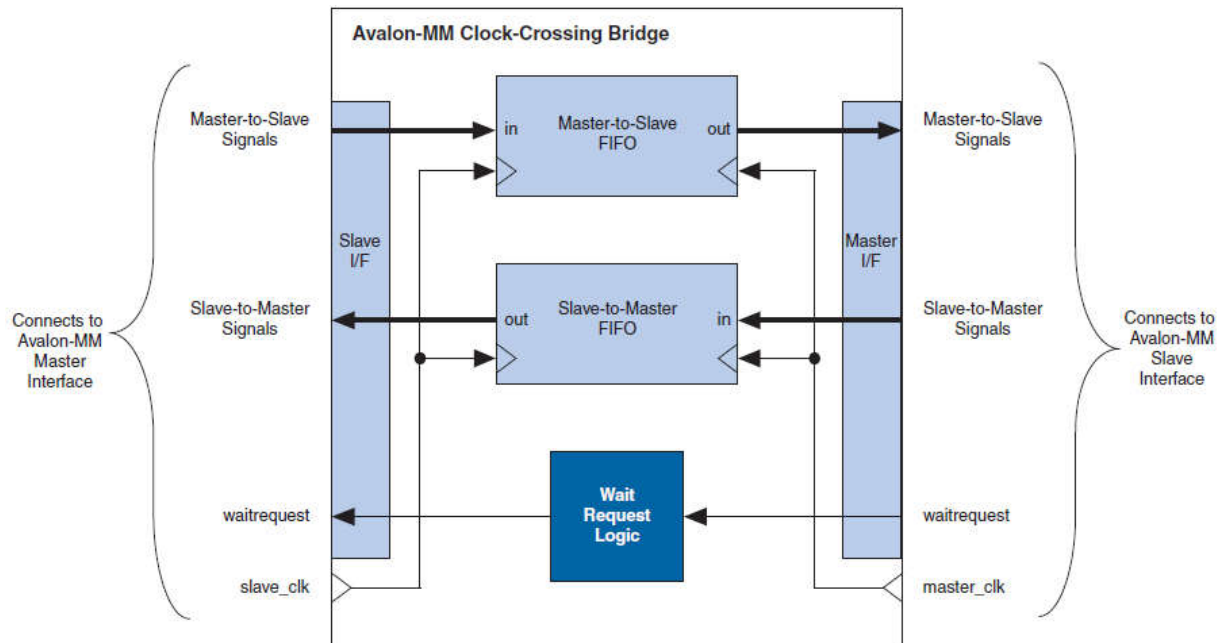
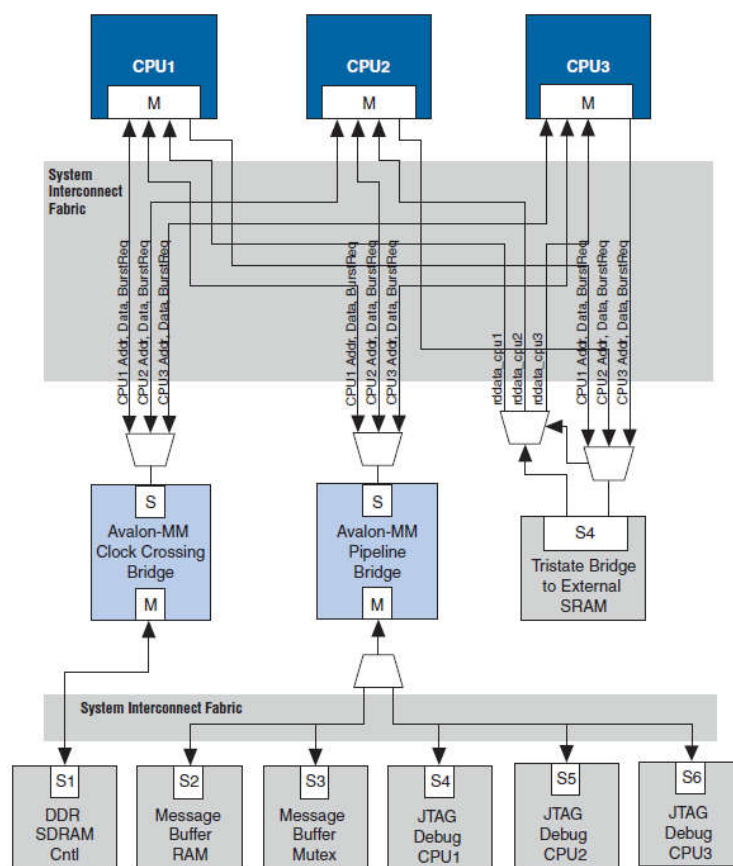
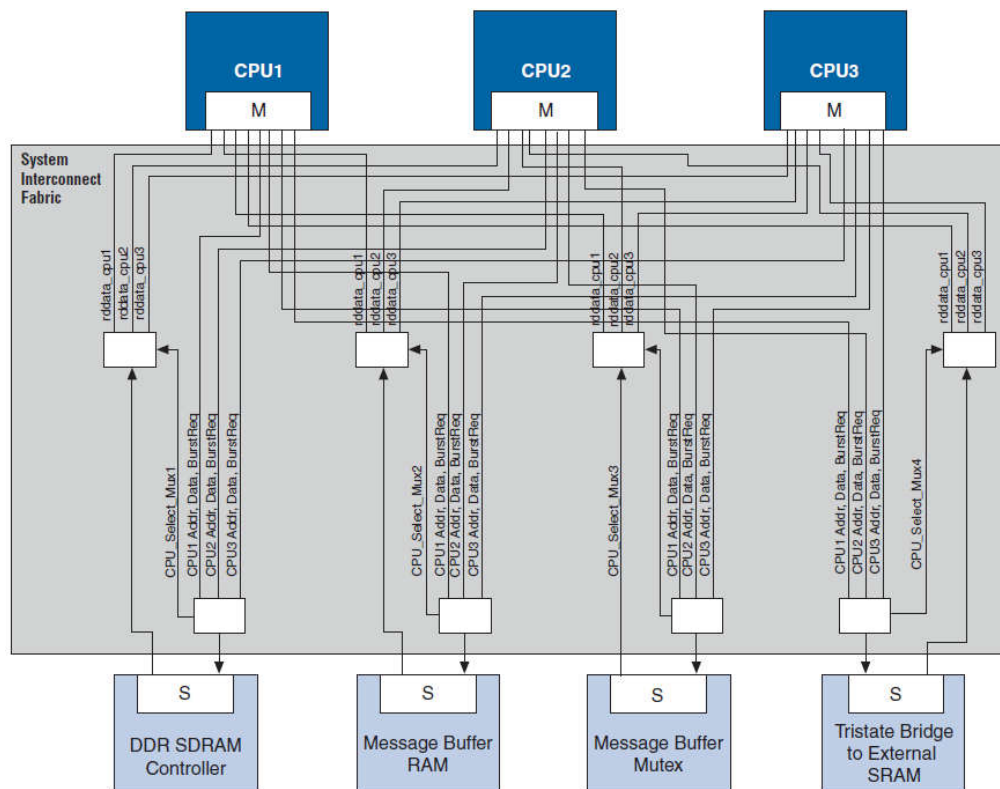


Figure 11–2. Example System Without Bridges—SOPC Builder View

Use	Connections	Module Name	Description	Base
☒		cpu1	Nios II Processor	
		instruction_master	Avalon Master	
		data_master	Avalon Master	
		jtag_debug_module	Avalon Slave	IRQ 0 0x02002800
☒		cpu2	Nios II Processor	
		instruction_master	Avalon Master	
		data_master	Avalon Master	
		jtag_debug_module	Avalon Slave	IRQ 0 0x00000800
☒		cpu3	Nios II Processor	
		instruction_master	Avalon Master	
		data_master	Avalon Master	
		jtag_debug_module	Avalon Slave	IRQ 0 0x00001000
☒		DDR_SDRAM_controller	DDR SDRAM High Performance Control...	
		s1	Avalon Slave	0x01000000
☒		message_buffer_RAM	On-Chip Memory (RAM or ROM)	
		s1	Avalon Slave	0x02001000
☒		message_buffer_mutex	Mutex	
		s1	Avalon Slave	0x02003000
☒		external_SSRAM_bus	Avalon-MM Tristate Bridge	
		avalon_slave	Avalon Slave	0x00000000
		tristate_master	Avalon Tristate Master	
☒		external_SSRAM	Cypress CY7C1380C SSRAM	
		s1	Avalon Tristate Slave	0xffffffff



❖ Insert:

- Clock Crossing Bridge
- Pipeline Bridge

Use	Connections	Module Name	Description	Clock	Base	End	IRQ
☒		cpu1	Nios II Processor				
		instruction_master	Avalon Master	clk			
		data_master	Avalon Master				
		jtag_debug_module	Avalon Slave		IRQ 0	IRQ 31	←x
☒		cpu2	Nios II Processor				
		instruction_master	Avalon Master	clk			
		data_master	Avalon Master		IRQ 0	IRQ 31	←x
		jtag_debug_module	Avalon Slave		0x03400800	0x03400fff	
☒		cpu3	Nios II Processor				
		instruction_master	Avalon Master	clk			
		data_master	Avalon Master		IRQ 0	IRQ 31	←x
		jtag_debug_module	Avalon Slave		0x03000800	0x03000fff	
☒		bridge	Avalon-MM Clock Crossing Bridge				
		s1	Avalon Slave	clk			
☒		ddr_sdram	DDR SDRAM Controller MegaCore Fun...				
		s1	Avalon Slave	clk	0x00000000	0x01ffffff	
☒		pipeline_bridge	Avalon-MM Pipeline Bridge				
		s1	Avalon Slave	clk	0x02000000	0x02001fff	
☒		message_buffer_ram	On-Chip Memory (RAM or ROM)				
		s1	Avalon Slave	clk	0x00000000	0x00000fff	
☒		message_buffer_mu...	Mutex				
		s1	Avalon Slave	clk	0x00001000	0x00001007	
☒		ext_ssram_bus	Avalon-MM Tristate Bridge				
		avalon_slave	Avalon Slave	clk			
		tristate_master	Avalon Tristate Master				
☒		ext_ssram	Cypress CY7C1380C SSRAM				
		s1	Avalon Tristate Slave	clk	0x03200000	0x033fffff	